Block practice · Differentiation and its applications

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Finding unknown constants in a curve from its gradient

Small changes and percentage changes

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Practice workbook · Finding unknown constants in a curve from its gradient

Using a known gradient (zero at a stationary point, or a tangent or normal gradient) and a known point to find letters such as p and q in the curve's equation.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-07

1. Question 4 · part a · 4 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part a only.

4 The curve C has equation $y = 2x^2 + px + q$ where p and q are integers.

The curve C has a stationary point at $(3, -5)$

(a) Show that $p = -12$ and find the value of q

(4)

(b) State, giving a reason, the nature of the stationary point.

(1)

(c) Find an equation of the normal to C at the point on C where $x = 1$

Give your answer in the form $ax + by + c = 0$

(6)

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P 6 6 0 2 7 R R A 0 1 2 3 6

1. Question 4 · part a · 4 marks · Working and answer

Working and answer

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2. Question 9 · part a · 5 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

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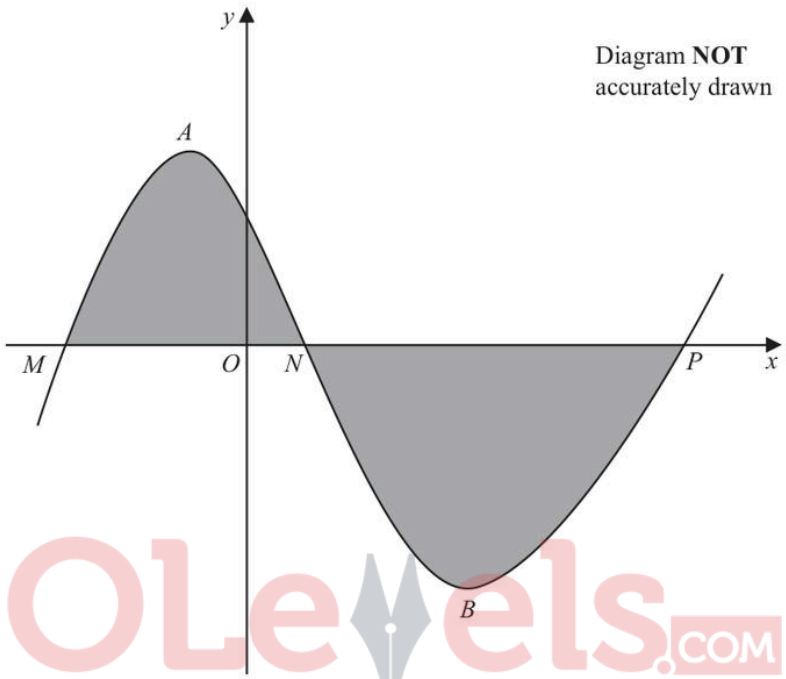


Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$ where

$$f(x) = 2x^3 + ax^2 + bx + c$$

The curve C has a maximum at the point A with coordinates $\left(-\frac{1}{3}, \frac{100}{27}\right)$ and a minimum at the point B with coordinates $(2, -9)$

Given that a, b and c are integers

(a) show that $a = -5, b = -4$ and $c = 3$

(5)

(b) (i) Show that $(x+1)$ is a factor of $f(x)$

(1)

(ii) Hence, or otherwise, use algebra to factorise $f(x)$ completely.

(3)

The curve C crosses the x-axis at the points M, N and P
The finite regions shown shaded in Figure 4 are bounded by the curve C and parts of the x-axis from M to N and from N to P

(c) Use algebraic integration to determine the total area of the shaded regions. Give your answer as an exact fraction.

(4)

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P 7 2 8 6 5 A 0 2 8 3 6

2. Question 9 · part a · 5 marks · Working and answer

Working and answer

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3. Question 3 · part a · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part a only.

3 A curve C has equation $y = \dfrac{ax - 3}{x + 5}$ where a is a constant and $x \neq -5$

The gradient of C at the point on the curve where $x = 2$ is $\dfrac{18}{49}$

(a) Show that $a = 3$

(3)

Hence

(b) write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis,

(2)

(c) find the coordinates of the point where C crosses

(i) the x-axis,

(ii) the y-axis.

(2)

(d) Sketch the curve C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

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3. Question 3 · part a · 3 marks · Working and answer

Working and answer

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4. Question 7 · part a · 6 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

7 The curve with equation $y = mx^2 + 64\sqrt{x} + 39$ has a stationary point with coordinates $(4, n)$ where m and n are integers.

Using calculus

(a) find the value of m and the value of n

(6)

(b) determine the nature of the stationary point.

(2)

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P 7 4 0 9 8 A 0 2 0 3 2

4. Question 7 · part a · 6 marks · Working and answer

Working and answer

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5. Question 9 · part a.i · 0 marks

2020 JAN · Paper 02

Further practice · Attempt part a.i only.

9 A curve C has equation

$$y = \frac{qx - 2}{x - p} \qquad x \neq p$$

The curve crosses the y-axis at the point A.

The line l with equation $y = x + 2$ is the normal to C at A.

(a) (i) Show that $p = 1$

(ii) Find the value of q.

(7)

(b) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(5)

The line l meets C again at the point D.

(c) Find the x coordinate of D.

(4)

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5. Question 9 · part a.i · 0 marks · Working and answer

Working and answer

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6. Question 9 · part a.ii · 7 marks

2020 JAN · Paper 02

Further practice · Attempt part a.ii only.

9 A curve C has equation

$$y = \frac{qx - 2}{x - p} \qquad x \neq p$$

The curve crosses the y-axis at the point A.

The line l with equation $y = x + 2$ is the normal to C at A.

(a) (i) Show that $p = 1$

(ii) Find the value of q.

(7)

(b) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(5)

The line l meets C again at the point D.

(c) Find the x coordinate of D.

(4)

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6. Question 9 · part a.ii · 7 marks · Working and answer

Working and answer

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7. Question 11 · part b.i · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part b.i only.

11 A curve C has equation

$$y = \frac{(2a-1)x+1}{ax-6} \qquad \text{where } a \text{ is a constant and } x \neq \frac{6}{a}$$

(a) Find $\frac{dy}{dx}$

(3)

The curve crosses the y-axis at the point A.

The normal to C at the point A is the line l with equation $66y - 72x + 11 = 0$

Show that

(b) (i) $a = 3$

(4)

(ii) the equation of C is $y = \frac{5x+1}{3x-6}$ where $x \neq 2$

(1)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes with their equations and the coordinates of the points where C crosses the coordinate axes.

(5)

The line l meets C again at the point D.

(d) Find the x coordinate of D.
Give your answer as an improper fraction.

(4)

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7. Question 11 · part b.i · 4 marks · Working and answer

Working and answer

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Practice workbook · Small changes and percentage changes

Using dy is roughly (dy/dx) times dx to estimate the change, or percentage change, in a quantity from a small change in the variable.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-13

1. Question 5 · part whole · 6 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part whole only.

5 y and x vary so that $y = xe^{-2x}$

Given that the value of x increases by 3%, use calculus to find, in terms of x, an estimate for the percentage change in y
Give your answer in the form $a(b - cx)$ where a, b and c are integers.

(6)

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1. Question 5 · part whole · 6 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part whole only.

Question 5 continued

(Total for Question 5 is 6 marks)

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1. Question 5 · part whole · 6 marks · Working and answer

Working and answer

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2. Question 5 · part whole · 5 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

5 A circle has radius $3r$ cm and area A cm^2

Given that the value of r increases by 0.05%

use calculus to find an estimate for the percentage increase in the value of A.

(5)

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2. Question 5 · part whole · 5 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 5 continued

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(Total for Question 5 is 5 marks)

2. Question 5 · part whole · 5 marks · Working and answer

Working and answer

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3. Question 8 · part a · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a only.

8 The volume of a sphere is $500\,cm^3$

(a) Calculate the radius, in cm to 3 significant figures, of the sphere.

(2)

The surface area of the sphere is increased by $20\,cm^2$

(b) Using calculus, find an estimate for the increase in the radius, in cm to 2 significant figures, of the sphere.

(5)

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3. Question 8 · part a · 2 marks · Working and answer

Working and answer

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4. Question 8 · part b · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.

8 The volume of a sphere is $500\,cm^3$

(a) Calculate the radius, in cm to 3 significant figures, of the sphere.

(2)

The surface area of the sphere is increased by $20\,cm^2$

(b) Using calculus, find an estimate for the increase in the radius, in cm to 2 significant figures, of the sphere.

(5)

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4. Question 8 · part b · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.



Question 8 continued

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(Total for Question 8 is 7 marks)

4. Question 8 · part b · 5 marks · Working and answer

Working and answer

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5. Question 7 · part a · 7 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

7 The equation of a curve is $y = \sqrt{\frac{e^{4x}}{2x-3}}$

When x is increased to $(x + \delta x)$, y increases to $(y + \delta y)$ where δx and δy are small.

(a) Show that $\delta y \approx \frac{e^{2x}(4x-7)}{(2x-3)^{\frac{3}{2}}}\delta x$

(7)

Given that $x = 2.5$

(b) find an estimate, to 2 significant figures, of the value of δy when the value of x increases by 0.2%

(3)

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5. Question 7 · part a · 7 marks · Working and answer

Working and answer

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6. Question 7 · part b · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

7 The equation of a curve is $y = \sqrt{\dfrac{e^{4x}}{2x-3}}$

When x is increased to $(x + \delta x)$, y increases to $(y + \delta y)$ where δx and δy are small.

(a) Show that $\delta y \approx \dfrac{e^{2x}(4x-7)}{(2x-3)^{\frac{3}{2}}}\,\delta x$

(7)

Given that $x = 2.5$

(b) find an estimate, to 2 significant figures, of the value of δy when the value of x increases by 0.2%

(3)

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6. Question 7 · part b · 3 marks

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Further practice · Attempt part b only.

Question 7 continued

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6. Question 7 · part b · 3 marks

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Further practice · Attempt part b only.

Question 7 continued

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6. Question 7 · part b · 3 marks

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Further practice · Attempt part b only.

Question 7 continued

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(Total for Question 7 is 10 marks)

6. Question 7 · part b · 3 marks · Working and answer

Working and answer

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7. Question 8 · part b · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

8 $y = \dfrac{2e^{3x+1}}{5x^2}$

(a) Find $\dfrac{dy}{dx}$

Give your answer in the form $\dfrac{Ae^{3x+1}(Bx-A)}{Cx^3}$ where A, B and C are prime numbers to be found.

(5)

The value of x increases by 2%

(b) Use your answer to part (a) to find an estimate, in terms of x, for the percentage change in y

Give your answer in the form $(Px-Q)$ where P and Q are integers.

(3)

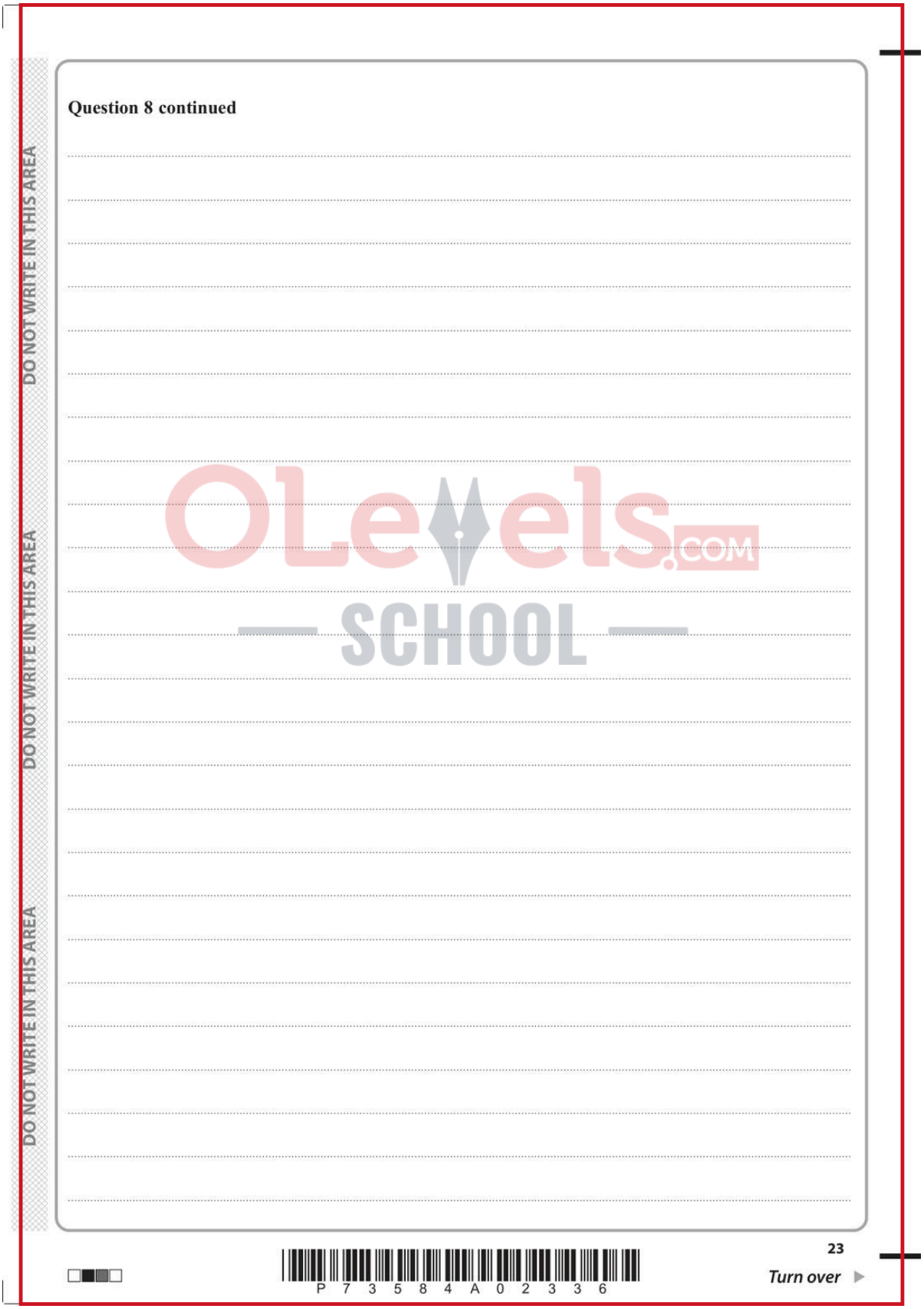
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7. Question 8 · part b · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.



7. Question 8 · part b · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

Question 8 continued

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7. Question 8 · part b · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

Question 8 continued

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(Total for Question 8 is 8 marks)

7. Question 8 · part b · 3 marks · Working and answer

Working and answer

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